3) **Deposit Creation through Lending:** When banks receive additional reserves, either through central bank injections or deposit inflows, they can expand their lending activities. Loans create new deposits, effectively increasing the money supply beyond the initial injection of base money.

4) **Multiplier Effect:** The expansion of bank loans and deposits sets off a multiplier effect as the newly created deposits become a part of the broader money supply. These deposits are re-deposited in banks or spent in the economy, leading to further rounds of lending and deposit creation.

5) **Limitations and Leakages:** The money multiplier process is subject to limitations and leakages that can dampen its impact on the money supply. These include excess reserve holdings by banks, currency drain (withdrawal of currency from banks), and non-bank financial intermediaries that do not fall under traditional reserve requirements.

Determinants of the Money Multiplier

Several factors influence the magnitude of the money multiplier and its effectiveness in expanding the money supply:

1) **Reserve Ratio:** The reserve requirement set by the central bank directly affects the money multiplier. A lower reserve ratio allows banks to lend out more of their deposits, leading to a higher multiplier effect.

2) **Currency Drain:** The propensity of the public to hold currency rather than deposits affects the effectiveness of the money multiplier. A higher currency drain reduces the deposit base on which banks can create new loans, thus lowering the multiplier.

3) **Excess Reserves:** The level of excess reserves held by banks influences their lending capacity. Higher excess reserves can limit the effectiveness of the multiplier process as banks may choose to hoard reserves rather than extend loans.

Implications for Monetary Policy and Economic Stability

Understanding the money multiplier process is crucial for central banks in formulating and implementing monetary policy. By adjusting the monetary base through open market operations, discount rate changes, or reserve requirement adjustments, central banks can influence the money supply and thereby regulate economic activity, inflation, and interest rates.

In conclusion, the money multiplier process serves as a cornerstone of monetary economics, providing insights into the mechanisms through which changes in the monetary base propagate through the financial system to impact the broader money supply. A nuanced understanding of the money multiplier process is essential for policymakers, economists, and students alike, as it underpins the formulation and implementation of effective monetary policy measures aimed at achieving macroeconomic stability and sustainable economic growth.

Check Your Progress 2

1) What is the role of the reserve requirement in the money multiplier process?

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2) How does the money multiplier process contribute to the expansion of the broader money supply?

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3) Why is understanding the money multiplier process crucial for central banks in formulating monetary policy?

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6.4 FACTORS AFFECTING THE MONEY MULTIPLIER AND HIGH POWERED MONEY

Understanding the dynamics of money supply is crucial for central banks and policymakers in managing monetary policy effectively. The money multiplier and high powered money are key concepts in this regard. The money multiplier represents the ratio of the money supply to the monetary base, while high powered money refers to the monetary base itself. Various factors influence both the money multiplier and high powered money, which in turn impact the overall money supply in an economy.

Factors Affecting the Money Multiplier:

1) **Reserve Requirements:** The reserve requirement ratio set by the central bank significantly affects the money multiplier. A lower reserve requirement leads to a higher money multiplier, as banks can lend out more of their deposits, increasing the money supply.

2) **Currency Drain:** When individuals hold more currency rather than depositing it in banks, the money multiplier decreases. This is because

currency held by the public does not contribute to the creation of new deposits through the banking system.

3) **Banking Behaviour:** The behaviour of banks also affects the money multiplier. If banks become more cautious and hold excess reserves rather than lending them out, the money multiplier decreases. Conversely, if banks are more aggressive in lending, the money multiplier increases.

4) **Central Bank Policy:** Monetary policy actions by the central bank, such as open market operations and changes in the discount rate, can influence the money multiplier. For example, when the central bank purchases government securities in open market operations, it injects reserves into the banking system, potentially increasing the money multiplier.

5) **Deposit Leakage:** Leakage of deposits from the banking system, such as through excess reserves, excess currency holdings, or withdrawals for purchases of securities or assets outside the banking system, reduces the effectiveness of the money multiplier.

Factors Affecting High Powered Money:

1) **Central Bank Operations:** High powered money is primarily influenced by the actions of the central bank. Central banks control the monetary base through various operations, including open market operations, changes in reserve requirements, and lending facilities.

2) **Government Transactions:** Government transactions, such as fiscal policy decisions that involve changes in government spending or taxation, can impact high powered money. For example, government expenditures financed by the central bank may increase the monetary base.

3) **Foreign Exchange Operations:** Foreign exchange interventions by central banks can affect high powered money, especially in economies with fixed or managed exchange rate regimes. Purchases or sales of foreign currency reserves impact the monetary base.

4) **Financial Crises:** During financial crises, central banks may engage in unconventional measures such as quantitative easing, which involves increasing the monetary base through large-scale purchases of financial assets.

5) **Banking Sector Dynamics:** Changes in the banking sector, such as the consolidation of banks or shifts in the composition of bank reserves, can influence high powered money.

The money multiplier and high powered money are interconnected concepts that play a crucial role in determining the overall money supply in an economy. Understanding the factors that influence these variables is essential for central banks and policymakers in formulating and implementing effective monetary policy. By monitoring and managing these factors, authorities can work to maintain price stability, promote economic growth, and achieve their policy objectives.

Check Your Progress 3

1) How does a decrease in reserve requirements affect the money multiplier?

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2) What impact does excess currency holdings by the public have on the money multiplier?

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3) Why is it important for central banks to consider government transactions when assessing high powered money?

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6.5 THE THEORY OF ENDOGENOUS MONEY SUPPLY

The theory of endogenous money supply challenges the traditional view of money creation, which suggests that the money supply is primarily determined by the actions of central banks. Instead, it posits that the money supply is largely endogenously determined by the demand for credit from borrowers and the ability of banks to create money through the process of lending. This theory has significant implications for understanding the dynamics of monetary policy, financial stability, and economic activity.

Key Concepts:

1) Credit Creation by Banks

In the endogenous money supply framework, banks play a pivotal role in money creation through the process of credit creation. When banks extend loans to borrowers, they simultaneously create new deposits in the accounts of the borrowers. These deposits effectively function as money in the economy, increasing the overall money supply.

2) Loanable Funds vs. Money Supply

Unlike the loanable funds theory, which suggests that banks act as intermediaries that lend out pre-existing deposits, the endogenous money supply theory emphasizes that banks create new money when they make loans. Thus, the money supply is not limited by the availability of pre-existing funds but is determined by the demand for credit and the lending capacity of banks.

3) Reserves and Central Bank Influence

While central banks retain some influence over the money supply through their control of the monetary base (reserves and currency), the endogenous money supply theory suggests that this influence is indirect. Central bank actions, such as setting interest rates or conducting open market operations, affect the cost of borrowing and the willingness of banks to extend credit, thereby influencing the money supply.

4) Creditworthiness and Economic Conditions

The ability of banks to create credit and expand the money supply is contingent upon the creditworthiness of borrowers and prevailing economic conditions. In times of economic expansion and confidence, banks may be more willing to lend, leading to an increase in the money supply. Conversely, during economic downturns or financial crises, banks may become more cautious, leading to a contraction in credit and the money supply.

5) Liquidity Preference and Monetary Policy Transmission

The endogenous money supply theory highlights the role of liquidity preference and interest rate expectations in monetary policy transmission. Changes in central bank policy rates influence the cost of borrowing and the demand for credit, thereby impacting the money supply and overall economic activity.

Graphical Representation of the Exogenous and Endogenous Money Supply Curves

We further discuss the difference between the exogenous and endogenous money supply curve with the help of Figure 6.1 whereby part A shows the exogenous supply curve and the factors affecting the shift in the exogenous money supply curve. Part B discusses the endogenous supply curve and the factors affecting the shift in the endogenous supply curve.

We discussed the following terms in Section 6.3:

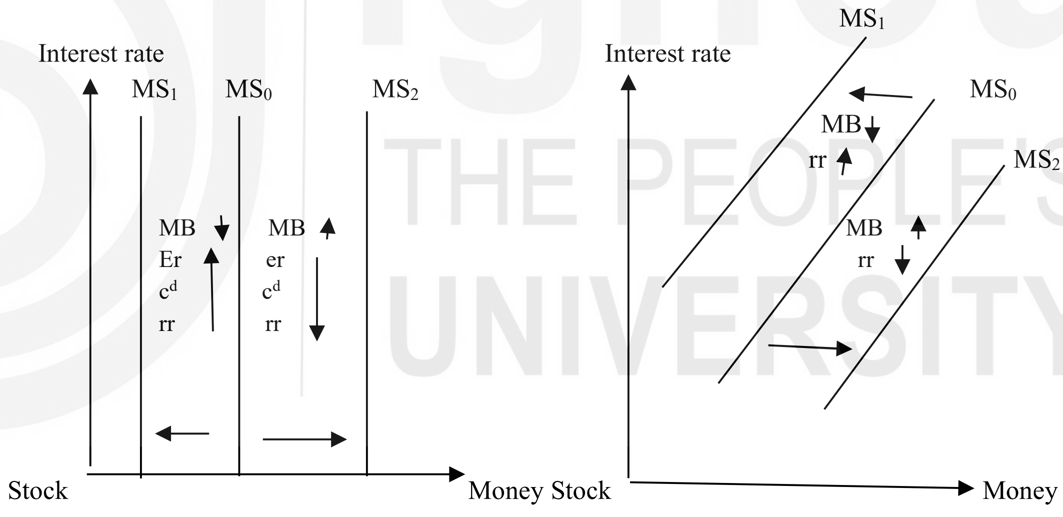
MB= Monetary Base; c^d = Currency to Deposit ratio; rr = required reserve to deposit ratio; er = excess reserves to deposit ratio; D = Demand deposit. These terms will be further used in the discussion.

Money supply curve shows the amount of money that suppliers are willing and able to supply at various interest rates in the economy. We know that there are four determinants of money supply, that is, MB, rr, er, and cd. From equation (10) from section 6.2, we know

Money supply=[c^d +1/ c^d +rr + er] * MB

But interest rates have no relation with MB and rr because it is determined by the Central Bank. Interest rates have impact on er and c^d. Economists differ in their views whereby some say that interest rate has no relation with er and c^d (Exogenous Money Supply Curve). Others say that interest rate has relation with er and c^d (Endogenous Money Supply).

In Figure 6.1 (A), we see that any change in the interest rates have no impact on the shape of the money supply curve, it is a vertical straight line (constant), say MS_0. As per The endogenous money supply curve in Figure 6.1 (B) we see that an increase in interest rates by the Central Bank Leads to more loans by the commercial banks in the market and hence increased money supply. So there is a positive relation between interest rates and money supply, upward sloping MS_0 curve in Figure 6.1 (B). At higher interest rates the customers prefer to deposit more money with the commercial banks to earn interest (c^d decreases), hence money available for loans increases with the commercial banks leading to increased money supply. Hence er and c^d impact money supply curve as per endogenous money supply surve.



A. Exogenous Money Supply Curve **B. Endogenous Money Supply Curve**

Fig. 6.1: Money Supply Curves and factors impacting shift in the curves

Further we discuss the shifts in both exogenous and endogenous money supply curve. In Figure 6.1 (A), we see that a decrease in MB and increase in er, c^d and rr leads to decrease in the money supply curve (leftwards), vice versa. Here it needs to be noticed that er and c^d are not impacted by interest rates, however, they do impact shifts in money supply.

In Figure 6.1 (B), we see that an increase in MB and rise in rr leads to increased money supply (rightward shift), MS_2 , vice versa. Here, er and c^d are impacted by interest rates and hence impact the money supply curve (upward sloping).

Implications:

1) Monetary Policy Effectiveness

Endogenous money supply theory suggests that monetary policy effectiveness may be influenced by the responsiveness of banks and borrowers to changes in central bank policy. Understanding the dynamics of credit creation and money supply determination is crucial for policymakers in assessing the impact of monetary policy on the economy.

2) Financial Stability Considerations

The theory of endogenous money supply emphasizes the interconnectedness of the banking system and the real economy. Financial stability considerations, such as the adequacy of bank capital and the management of systemic risks, are essential for ensuring the smooth functioning of the endogenous money supply mechanism.

3) Role of Central Banks

While central banks retain a significant role in influencing monetary conditions and financial stability, the endogenous money supply theory underscores the importance of understanding the behaviour of banks and borrowers in shaping the overall money supply dynamics.

The theory of endogenous money supply provides a different perspective on the process of money creation and the role of banks in the economy. By highlighting the endogenous nature of the money supply and the importance of credit creation by banks, this theory deepens our understanding of monetary policy transmission, financial stability, and the functioning of the monetary system. Incorporating these insights into economic analysis and policymaking can lead to more effective strategies for managing monetary conditions and promoting sustainable economic growth.

Check Your Progress 4

1) How does the theory of endogenous money supply differ from the traditional view of money creation?

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2) How do changes in central bank policy rates influence the money supply according to the theory of endogenous money supply?

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6.6 LET US SUM UP

The foundation of any economy lies in its monetary framework. Understanding the nuances of money supply is imperative for policymakers and economists alike. At the core of this understanding is high-powered money, which forms the basis for the broader monetary aggregates. Central banks wield significant control over the money supply through various tools at their disposal. By manipulating the monetary base through operations such as open market transactions, reserve requirement adjustments, and changes in the discount rate, central banks can influence economic activity, inflation, and interest rates.

The money multiplier process illustrates how changes in the monetary base trickle down through the financial system, amplifying the overall money supply. This process begins with an injection of base money into the banking system, leading to the creation of deposits as commercial banks extend loans. The subsequent expansion of bank loans and deposits triggers a multiplier effect, resulting in a proportional increase in the money supply. However, the effectiveness of this process is subject to various factors, including reserve requirements, currency drain, and the level of excess reserves held by banks.

In contrast to traditional views of money creation, the theory of endogenous money supply sheds light on the crucial role of banks in determining the money supply through credit creation. According to this theory, money is generated when banks extend loans to borrowers, thereby contributing to the expansion of the money stock. Understanding the intricacies of credit creation and the behaviour of banks and borrowers is essential in shaping our comprehension of money supply dynamics.

Overall, this exploration provides a comprehensive overview of the mechanisms governing money supply, highlighting the roles of high-powered money, the money multiplier process, and the theory of endogenous money supply. By delving into these concepts and their implications for monetary policy and economic stability, stakeholders can deepen their understanding of macroeconomic phenomena and make informed decisions to steer economies towards sustainable growth and stability.

6.7 KEY WORDS

High-Powered Money : The total amount of currency in circulation and reserves held by the central bank, serving as the foundation for the broader money supply in an economy.

Money Multiplier : A ratio representing the change in the money supply relative to the change in the monetary base, illustrating how an initial injection of high-powered money leads to a multiple expansion of the broader money supply through bank lending and deposit creation.

Exogenous : External or outside influence; in the context of money supply, exogenous refers to factors that are determined externally, such as by central banks.

Endogenous Money Supply : A theory challenging the traditional view of money creation, suggesting that the money supply is largely determined by the demand for credit from borrowers and the ability of banks to create money through lending, rather than by central bank actions alone.

Monetary Base : Synonymous with high-powered money, it represents the total amount of currency in circulation and reserves held by the central bank.

Open Market Operations : Actions undertaken by central banks to buy or sell government securities in the open market, influencing the amount of reserves in the banking system and thereby impacting the money supply.

Reserve Requirements : Mandated reserves that commercial banks are required to hold by central banks, affecting the amount of excess reserves available for lending and thus influencing the money supply.

Liquidity Preference : The desire of individuals and businesses to hold assets in liquid form, such as cash or demand deposits, rather than in illiquid forms such as physical assets or long-term investments.

6.8 SOME USEFUL BOOKS

Errol D'Souza (2012) Macroeconomics 2nd edition Pearson Education, New Delhi

McCallum, Benett Monetary Economics

6.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1. Currency and reserves.

2. Through open market operations, the central bank buys or sells government securities in the open market.

Check Your Progress 2

1. Reserve requirement acts as a constraint on the amount of money banks can create through lending.

2. Refer to Section 6.3

3. Refer to Section 6.3

Check Your Progress 3

1. A lower reserve requirement leads to a higher money multiplier, as banks can lend out more of their deposits, increasing the money supply.

2. When individuals hold more currency rather than depositing it in banks, the money multiplier decreases. This is because currency held by the public does not contribute to the creation of new deposits through the banking system.

3. Government transactions, such as fiscal policy decisions that involve changes in government spending or taxation, can impact high powered money.

Check Your Progress 4

1. Refer to Section 6.5

   While central banks retain some influence over the money supply through their control of the monetary base (reserves and currency), the endogenous money supply theory suggests that this influence is indirect. Central bank actions, such as setting interest rates or conducting open market operations, affect the cost of borrowing and the willingness of banks to extend credit, thereby influencing the money supply.